Uchaaran

A little practice. A deeper connection.

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The problem: helping children learn Gurbani

Children learning Gurbani need both a reason to return to daily practice and guidance to give each word the attention it deserves. Parents want to nurture that connection, but consistent access to an ustaad or time to guide every recitation is not always available. Conversations with Khalis Foundation helped us focus on two priorities: sustaining engagement and supporting pronunciation and understanding. We set out to make practice inviting and feedback clear, with the ustaad's recitation as the standard and rewards tied to learning rather than screen time.

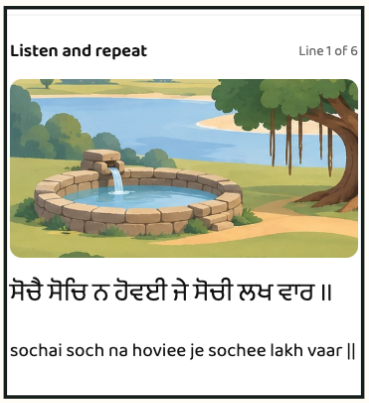
Two modes. One path toward mastery.

1. Practice at your pace

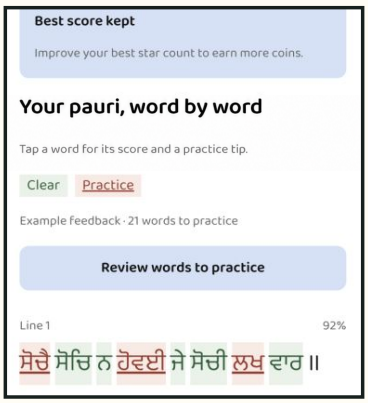
Listen and recall with Gurmukhi, Roman-letter Punjabi and English meaning. Revisit a line or word.

2. Take the pauri challenge

Recite a line and tap Done. Review word feedback. Earn stars and Nanak coins for character rewards.



Listen + recall



Review each word

What changed since midterm

Organizers asked for more attainable goals and simpler feedback for students. We added line-by-line scoring, short prompts, and an ML maintenance guide for Khalis Foundation. We also compared ML models to find the best fit for Gurbani pronunciation, focusing on meaningful word-level scoring and clear, actionable guidance for students.

Why a memory palace?

For grades 6 to 8, familiar scenes link each line to a place along 38 pauris. Sikh-inspired journeys invite recall and reflection. The ustaad guides pronunciation; rewards follow learning, not screen time.

Rooted in the Panth

Helping the next generation build a lasting connection with Gurbani through careful pronunciation, understanding, and daily practice. Playful technology supports that tradition, with Gurmukhi and the ustaad's guidance at the center.

Private processing, by design

Recordings are processed privately in Azure. Temporary server recordings are deleted after scoring, with no use for public-model training.

Khalis Foundation can host the model in its own shared AWS or Azure environment. On-device packaging is another option for fully local processing that keeps recordings on device.

Tools and implementation

React Native · JavaScript · Azure-hosted IndicConformer · BaniDB · reference audio. Codex and Paper supported development and design.

Watch demo ↗ | **Try the app ↗** | **View code ↗** | **Design process ↗**

Ready to help Khalis Foundation launch. An implementation PR is open in the main app repository for approval and deployment! We would love to help with infrastructure and support your technical team.